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Welcome to the Dynamic Planet Division C Invitational Contest!

There are **70 questions** for a total of **146 points**!

Make sure to read all questions and background information **thoroughly** and answer the questions. Calculation questions are generally given about a 2% error margin for getting correct, that is, as long as your answer is within roughly 2% of the accepted answer on the answer key, your answer will be deemed correct. Significant figures are not graded, so use as many you feel comfortable with.

Do **not** write on this document - an answer document will also be given to you.

Enjoy the contest!

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Crater lake is a large and circular lake in Oregon. It sits atop a caldera from a recent eruption and collapse of the volcanic mount Mazama. As a result, some low-end geothermal activity at the bottom of the lake is a source of some nutrients. Its level, though subject to large seasonal variations of rainfall, is largely contained within a range of 60 cm. Answer the questions below. See figure 1 for guidance on some of these questions.

1. (**8pts**) Fill the table summarizing the water flow-through characteristics of Crater lake. Note that evaporation makes up 28% of the annual water losses of the lake, and outflow makes up the remainder. The lake also covers 78.5% of its total watershed area. Precipitation and evaporation are measured akin to that seen in rain gauges - units of depth. Finally, precipitation and evaporation are in their own categories separate from lake inflow and outflow.

2. (**3pts**) Due to a lack of surface outflow, Crater lake would be a(n) (a.) _____ lake except that the lake loses water through underground seepage and supplies springs that drain away from the lake. Thus, Crater lake is actually a(n) (b.) _____ lake. *Write the letter to the corresponding answer for each blank.*

a. Exorheic
b. Lacustrine
c. Perennial
d. Dry
e. Endorheic
f. Meromictic
g. Cryptorheic
h. Oligotrophic

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3. (**6pts**) Considering that all of Crater lake's outflow is underground, Darcy's law can be used to characterize the flow rate. Calculate the speed of the water flow through the ground in mm/s 3km away from the lake edge. Assume a constant level of aquitard and water head drop of 80m, as well as a sediment porosity of 25%. Then calculate the permeability of sediment in darcy. To get started, assume the lake is a cylinder, with its bottom being the level of the aquitard, and lake radius calculable from a circular surface area. Recall that water produces a pressure of 9,800 Pa per meter of height and has a viscosity of 0.001Pa•s, and that a darcy is 9.869x10^-13m^2.

4. (**2pts**) Beyond all of the things I told you to assume, what is one big assumption you made, and why is it a problem? Consider the topographic map, and that each red grid square is roughly 1.5km.

5. (**1pt**) What direction is Sand creek flowing at its point near Anderson spring and Mount Scott?

a. Northeast
b. Southeast
c. Southwest
d. Northwest

6. (**4pts**) The nearby Bear creek seems to angled in the same direction as Sand creek. Is Bear creek flowing in the same direction as Sand creek? If not, what direction? How can you tell? Mention two ways this is figured out.

7. (**1pt**) What kind of drainage pattern do the streams around/within 1 radius of Crater lake have?

a. Dendritic
b. Annular
c. Radial
d. Trellis
e. Rectangular

8. (**1.5pts**) A bit further off on the left side of the topographic map, the streams seem to start angling into a different drainage pattern. What is its name?

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9. (**1pt**) Crater lake is known for its very clear deep blue water. Crater lake can be thus classified as a(n):

   a. Oligotrophic lake
   b. Limnic lake
   c. Mesotrophic Lake
   d. Eutrophic lake
   e. Holomictic lake

10. (**5pts**) Crater lake is also known for being a lake that mixes twice a year. Complete the lake surface temperature chart, depicting lake mixing in the late spring (April-May) and early winter (November-January). The maximum annual temperature of 14ºC and minimum annual temperature of 2ºC are already placed into the chart. Temperatures do not need to be perfect/exact, but they will be graded on how realistic they are. Answer in whole ºC.

11. (**1pt**) What is the biggest impact that heavy snow over the winter would have on the mixing of Crater lake?

   a. Heavy snow would cool down the lake surface, and create an area of really cold and less dense water at the surface, ending mixing early in the winter.
   b. Heavy snow would create a big snowpack, which would melt in the spring, providing cold water to keep lake surfaces cool, extending mixing in the summer.
   c. Heavy snow would increase the surface mass balance of the lake, which would promote mixing to redistribute the mass, extending mixing in the winter.
   d. Heavy snow would create a big snowpack, which would melt in the spring, providing cold water that would immediately sink in the quickly warming surface water to the cold water below, strengthening the thermocline and ending mixing early in the summer.

12. (**5pts**) During its periods of mixing, algae is found in an even distribution throughout the upper 300m of Crater lake's water column. However, during the summer period of stratification, algae are primarily 50-100m below the lake surface. Suggest why this is the case.

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13. (**1pt**) On some of the streams on the Crater lake topographic map, there appears to be circles on the close points of the streams to the lake. What do these circles represent on the topographic map? Choose the best answer.

a. Artesian Wells
b. Caves
c. Disappearing Streams
d. Springs
e. River Sources

14. (**1pt**) Choose the topographic map feature that would have water moving in the opposite manner to the one discussed in the previous question.

a. 
b. 
c. 
d. 

Lake Okeechobee is a large freshwater lake in Florida, having a surface area of 1900 square kilometers, but only an average depth of 2.7 meters. In 1928, a powerful hurricane passed over the region and caused at least 2,500 deaths over all of florida - the majority of them being along the lake shoreline. Many deaths occured when the waters of the lake breached the surrounding 2-meter flood dike and flooded homes as the storm's eye approached and caused flooding on the southern shores.

15. (**2pts**) What could have contributed significantly to the flooding from lake Okeechobee during the hurricane? Select all correct options.

a. Heavy rain in the days preceding the hurricane
b. Seiche
c. Storm surge
d. Heavy rain as the hurricane passed over

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16. (**3pts**) One of the reasons why the hurricane was so problematic is that it managed to also flood the north shores *immediately* after the eye of the hurricane passed over the lake. Was the cause of the flood on the northern shore different than that of the southern shore? If so, what caused it? If the cause did *not* change, why?

17. (**2pts**) A layer of clay underlays lake Okeechobee. In the same vicinity, however, is the Floridan aquifer, which approaches the lake from the underside. Choose the correct answers below about this system of two freshwater bodies.

a. The Floridan aquifer supplies lake Okeechobee
b. The Floridan aquifer is unconfined aquifer at lake Okeechobee
c. The Floridan aquifer is confined at lake Okeechobee
d. Lake Okeechobee is part of an unconfined aquifer
e. Lake Okeechobee is part of a confined aquifer
f. Lake Okeechobee supplies the Floridan aquifer

18. (**1.5pts**) Lake Okeechobee is fairly well-mixed at all times of the year. What term describes the behavior of this lake?

a. Meromictic
b. Monomictic
c. Dimictic
d. Oligomictic
e. Polymictic
f. Omnimictic

19. (**3pts**) Answer these questions discussing the mixing of Lake Okeechobee.

a. Briefly, why is Lake Okeechobee so well-mixed all year?

b. Occasionally, stratification does occur on lake Okeechobee. Briefly, what conditions would allow this?

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20. (**1.5pts**) Look at figure 2. What are the two types of glacial features seen in the image?
    a. A pingo on the top right, and an esker running down along the middle
    b. A kettle lake on the top right, and a pingo running down along the middle
    c. An esker on the top right, and a pingo running down along the middle
    d. A pingo on the top right, and a kettle lake running down along the middle
    e. A kettle lake on the top right, and an esker running down along the middle
    f. An esker on the top right, and a kettle lake running down along the middle

21. (**2pts**) Calculate the approximate sinuosity of the wiggly feature in figure 2.

22. (**1pt**) How sorted is the soil in this area?
    a. Well-sorted
    b. Poorly-sorted
    c. Moderately sorted
    d. Unsorted
    e. Not classifiable; this is bedrock

23. (**1pt**) How permeable is the soil in this area? Why?
    a. Relatively impermeable, because the soil can hold a lake on top of it without losing it to the ground
    b. Relatively impermeable, due to the sortedness of the soil
    c. Permeable, due to the sortedness of the soil
    d. Permeable, because water can flow from underground to support lakes

24. (**1pt**) An explorer comes across a feature as seen in figure 3. What type of structure did this feature form from?
    a. Drumlin
    b. Volcano
    c. Lake
    d. Native American burial mound
    e. Pingo

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25. (**1pt**) There are different versions of the structure identified in the previous question that could form this feature. What version was it?
    a. Hydrostatic
    b. Composite
    c. Hydraulic
    d. Not enough information

26. (**1pt**) True or False? Ice caps and glaciers contain more freshwater than groundwater.
    a. True
    b. False

27. (**1pt**) You assign Benny to survey a river and run some analyses on it. He later comes back with you to report his results. In one of his surveys, Benny reports that a stream has a sinuosity of 0.85. Interpret that result.
    a. Benny is wrong, and a dumb stupid idiot that should not be a hydrologist, and should be fired on spot… there is NO WAY ON EARTH a stream can have a sinuosity of less than 1. What a waste of your time and money…
    b. The stream is a very meandering river, and likely has a shallow angle of descent as it approaches the ocean.
    c. The stream is a mildly twisty river, and likely is still a fair distance away from an ocean and its coastal plains.
    d. The stream is a very straight river, and is likely a stream that is confined to a fault on bedrock, and thus would be part of a rectangular drainage system if you zoom out.

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28. (**1pt**) Since teamwork is an important part of science and surveying, you also had Arianna on the case with Benny to analyze the stream. She independently reported some results to you, including sinuosity, which she reports being 1.18. Interpret that result.

a. Arianna is wrong, and a dumb stupid idiot that should not be a hydrologist, and should be fired on spot… there is NO WAY ON EARTH a stream can have a sinuosity of greater than 1. What a waste of your time and money…
b. The stream is a very meandering river, and likely has a shallow angle of descent as it approaches the ocean.
c. The stream is a mildly twisty river, and likely is still a fair distance away from an ocean and its coastal plains.
d. The stream is a very straight river, and is likely a stream that is confined to a fault on bedrock, and thus would be part of a rectangular drainage system if you zoom out.

29. (**4pts** ***Tiebreaker 1***) Okay, there are two different values reported for the sinuosity of this stream. At least one of them has to be wrong, or maybe they used different methodology. Write down the correct formula to calculate sinuosity, and if you believe one of them used an incorrect calculation, write down the formula that the incorrect person used, and *leave the methodology section blank or crossed out.* If using variables, define what each one is! If you think it is a methodology difference, suggest why the two values are different, and *leave the incorrect formula section blank or crossed out.*

Correct:

Incorrect:

Methodology Difference:

30. (**2pts**) It is your turn. Calculate the sinuosity of the creek that passes near Llaos Highway in figure 1.

31. (**1pt**) On the same river of your analysis, your team reports high lead content in the water. Where is this in the erosion model of a stream?

a. Bed load
b. Dissolved load
c. Suspended load
d. Wash load

32. (**2pts**) What likely caused the high lead levels? Circle two correct answers
    a. Nothing other than the high mineral content in mountainous areas; this relatively common and natural, so it is not pollution
    b. Mine runoff; this mostly is point-source pollution
    c. Mine runoff; this mostly is nonpoint source pollution
    d. Agricultural runoff; this mostly is nonpoint-source pollution
    e. Agricultural runoff; this mostly is point-source pollution
    f. A large boat carrying lead spilled its contents in a tragic disaster. This is accidental pollution.

33. (**3pts**) Listed below are some other characteristics this stream could have. Circle the letters commonly associated with the stream hypothetically (may or may not be realistic based on the previous question) having a mine drainage issue.
    a. High acidity
    b. High alkalinity/basicity
    c. Hypoxic conditions
    d. High sulfate concentrations
    e. High nitrate concentrations
    f. High phosphate concentrations
    g. High pH
    h. High concentrations of other metals

34. (**3pts**) By the point a certain river meets the ocean, it has had a total of 27 tributaries/sources. What is the stream order according to each system? If more than one value is impossible, write a range from the minimum possible value to the maximum possible value.
Shreve: _____ Strahler: _____ Classical: _____

35. (**4pts**) Just before meeting the ocean, the river is 48 meters wide and 7 meters deep. The river dropped 15 meters on the 20 kilometers before reaching the ocean. Treating the river channel as an isosceles triangle (with its widest angle being the bottom point of the river) in cross-section, calculate the flow velocity and discharge rate of the river using Manning's equation. Assume a Manning roughness of .030.

36. (**1pt** ***Tiebreaker 2***) Assume that the river is now flooding. Assuming the same channel shape, proportions, and characteristics, how much will the flow (both speed and volume) of the river change if the waters flooded 10% above their usual levels (70cm above normal)?

a. Flow speed increases by 10.0%, and discharge rate by 33.1%
b. Flow speed increases by 6.56%, and discharge rate by 28.9%
c. Flow speed increases by 6.56%, and discharge rate by 17.2%
d. Flow speed increases by 10.0%, and discharge rate by 21.0%

37. (**2pts**) Oftentimes, when major rivers flood, their flow velocities *decrease* rather than increase. Why is that the case? Choose the correct choices.

a. The river spills to flood its floodplain, which typically is flatter than the riverbed, which lowers the rate of descent, or friction slope, of the river, which slows down flow velocity
b. The river spills to flood its floodplain, which typically has a higher Manning's roughness due to its vegetation, which slows down flow velocity
c. The river spills to flood its floodplain, which makes it cover a much larger surface area per volume (or much larger perimeter per cross-section area), which decreases its hydraulic radius and slows down flow velocity
d. The river water levels rise, which effectively acts to convert some of the kinetic energy of the flow velocity into gravitational potential energy, slowing down the river's flow velocity
e. The river water levels rise, which causes the friction slope, or rate of descent, to decrease, which then causes the flow velocity to decrease.

38. (**1pt**) It isn't a real *Dynamic Planet* exam until I request you to run a *second* Manning equation calculation! A river at some point in its course has a profile akin to that of what is in figure 4. It has a flow velocity of 80cm/s and a flow rate of 9600m^3/s. Assume a manning roughness of 0.35, calculate the rate of descent of the river per kilometer. I'll be nice and make it multiple choice.

a. 87.6 meters per kilometer
b. 7.67 meters per kilometer
c. 49.0 meters per kilometer
d. 2.40 meters per kilometer

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39. (**3pts**) Name the features labeled in figure 4.
    A. _____
    B. _____
    C. _____

40. (**1pt**) Which way is the river curving towards based on this cross-section?
    a. To the right!
    b. To the left!
    c. This is a trick question! The river is meandering, but it is impossible to know which way it is curving!
    d. This is a trick question! The river is not meandering, so there is a curve in the river at this spot as it is going straight!

41. (**1pt**) A fair bit upstream, the feature marked as "a" in figure 4 was more pronounced and in multiples at spots, seemingly dividing the river into multiple channels. What is this called?
    a. A delta
    b. A floodplain
    c. A braided channel
    d. An alluvial fan
    e. A meandering stream

42. (**2pts**) What can be said about this stream as it passes through the area mentioned in the previous question? Choose all that apply.
    a. It has a very slow flow and slow rate of descent
    b. It has high sinuosity and produces features such as oxbow lakes
    c. It has a moderate flow rate with a somewhat steep rate of descent
    d. It carries a small amount of fine sediment
    e. It carries coarse sediment in fairly large amounts

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43. (**1pt**) In figure 4, which side of the river is there sediment depositing? At what side of the river is sediment being weathered?

a. At the left side (point C), sediment is being deposited; on the right side (point B), sediment is being weathered
b. At the right side (point B), sediment is being deposited; on the left side (point C) sediment is being weathered
c. At the left side (point C), sediment is being both weathered and deposited
d. At the right side (point B), sediment is being both weathered and deposited

44. (**2pts**) On the portion of the river where sediment is being deposited, what can be said about the deposits? Select all correct choices.

a. The deposits are well-sorted
b. The deposits are poorly-sorted
c. Some or all of the deposits come from the bed load
d. Some or all of the deposits come from the suspended load
e. Some or all of the deposits come from the dissolved load

45. (**1pt**) What else can be said about these depositional features as a whole?

a. They are composed of all fine particles such as clay, silt, and maybe sand.
b. They are composed of all coarse particles such as gravel and pebbles.
c. They are composed of coarse particles nearest to the thalweg (central part with the fastest flow) of the channel, and fine particles nearest to the edge of the channel
d. They are composed of coarse particles in the upstream-facing portion, and of fine particles in the downstream-facing portion
e. All of these statements are false

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46. (**1pt**) How are these types of erosional and depositional features formed? Choose the *best* answer.

a. A helicoidal secondary flow circulation is associated with a river meander that pushes sediment from the outside of the curve to the inside.
b. A helicoidal secondary flow circulation is associated with a river meander that pushes sediment from the inside of the curve to the outside.
c. Water flows fastest on the outside of the river to save inertia, maximizing erosion, while water flows slowest on the inside of the river, maximizing deposition.
d. Water flows fastest on the inside of the river to save inertia, maximizing erosion, while water flows fastest on the inside of the river, maximizing deposition.

47. (**6pts**) A certain area has identical streams with identical watersheds and landforms. Stream A flows through land that has been recently clear-cut, while stream B flows through land that has not been disturbed. Both streams supply the local groundwater, as well as merge (as tributaries) into a larger stream. Assume that a rainstorm overspreads the region and dumps an equal amount of heavy rain over both streams' watersheds over a short amount of time. Construct a qualitative storm hydrograph of both streams, starting from a bit before the storm, and ending when both streams get back to normal flow level. Both graphs should be *clearly* labeled and overlaid (plotted on the same graph/pair of axes). A pair of axes is given on the answer document. Make sure to give an appropriate label to the axes, as well as the entire chart.

48. (**1pt**) Over the *entire* period from before the storm to after water levels return to normal, which stream would discharge more excessive flow?

a. Stream A
b. Stream B
c. Neither; they both discharge the same excess water, just over differing time periods
d. Neither; they both discharge the same excess water over the same time period
e. Not enough information

49. (**1pt**) Which stream exhibits more downcutting? Why?
    a. Stream A because stream B would carry more sediment which would yield some deposition, which would counteract any downcutting.
    b. Stream B because stream A would carry more sediment which would yield some deposition, which would counteract any downcutting.
    c. Stream A because stream A would carry more sediment which would increase its erosional power via abrasion, which would enhance downcutting.
    d. Stream B because stream B would carry more sediment which would increase its erosional power via abrasion, which would enhance downcutting.

50. (**1.5pts**) Looking at a water table level map, which direction would the height contour "v's" associated with each river point toward?
    a. Downstream for both rivers
    b. Downstream for A, upstream for B
    c. Upstream for A, downstream for B
    d. Upstream for both rivers
    e. Perpendicular to both rivers
    f. There will be no "v" notch in the height contours of the water table level associated with the rivers

51. (**1.5pts**) Suppose the typical stage of a river is 10 meters, and its 200-year flood stage is 22 meters. What would its 100-year flood stage be?
    a. 10 meters
    b. 12 meters
    c. 16 meters
    d. 18 meters
    e. 22 meters
    f. Correct answer not given; the flood stage would be _____ meters (fill in the blank)
    g. Impossible to know without more information.

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52. (**6pts**) Answer these questions about living on the edge of a 200-year floodplain. Answers have to be exact (no 2% leniency is given here)!

a. What is the yearly % chance of having a flood?

b. What is the % chance of NOT having flooded after 50 years?

c. How many full years does it take to have at least a 50% chance of being flooded?

d. What are the % chances of getting flooded 3 years in a row?

53. (**3pts**) Select all answer choices that are true to alluvial fans.

a. They are the same as river deltas
b. They have well-sorted sediment, especially near the the edge of the fan
c. They indicate a river dropping rapidly to its base level over the course of the fan
d. There are two types: fluvial, which is dominated by intermittent or continuous overloaded stream flow; and debris flow, which is dominated by occasional debris flows
e. The slope of an alluvial fan typically gets steeper as it spreads out
f. They typically occur when a stream exits a mountainous region to a flatter region
g. The image in figure 5 depicts an alluvial fan

54. (**3pts** *__Tiebreaker 3__*) Note the little incisions into the hill next to the main feature (to the immediate left and right of the collapsed part). What are these called? Briefly, what is their cause?

55. (**1pt**) A river flows through a plain underlain by impermeable bedrock. What is the local/structural base-level of the river?

a. The river surface/bank
b. The top of the bedrock
c. Deep below the bedrock
d. Sea level

56. (**1pt**) Uplift occurs on the bedrock over portions of this river. What happens in the short term?

a. The stream uplifts as well
b. The stream redirects to flow down away from the uplifted region
c. The stream becomes incised into the uplifted region
d. The stream carves a broad v-shaped valley

57. (**1pt**) The river drains into an endorheic basin. What would be the river's ultimate base level?

a. Whatever level the bottom of the basin is at
b. Whatever level the water level of the basin is at
c. Sea level
d. Above sea level
e. Below sea level

58. (**2pts**) Select all examples of knickpoints.

a. Rapids
b. Terraces
c. Waterfalls
d. Valleys
e. Deltas

Chesapeake Bay is a large estuary of the mid Atlantic states of the US. View figure 6 for an overview of the hydrogeology of Chesapeake bay. Answer the following questions about Chesapeake bay.

59. (**1.5pts**) What drainage pattern does the upper Potomac river have?

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60. (**1.5pts**) Which two points share the most similar drainage basin?
    a. A and B
    b. A and C
    c. A and D
    d. B and C
    e. B and D
    f. C and D

61. (**1pt**) A lot of water in Chesapeake bay is brackish. What does brackish mean?
    a. The water is part of a wetlands system and is murky
    b. The water has a salinity between that of fresh water and sea water
    c. The water has high nutrient content to support life
    d. The water is hypersaline (more salty than sea water)

62. (**5pts**) 15,000 years ago, there used to be a river valley where most of Chesapeake bay now covers. What changed to make Chesapeake bay the way it is today - and what type of process occurred? Fill in the blanks with relevant vocabulary, phrases, and/or events to explain.

Roughly 10,000 years ago (a.) _____. This resulted in (b.) _____, which caused the ultimate (c.) _____ of the ancestral Chesapeake bay river to (d.) _____. This caused the valley to (e.) _____ and form a feature known as a ria inlet.

63. (**2pts**) What is a rock type that is common with Karst? Select all correct answers.
    a. Granite
    b. Andesite
    c. Sandstone
    d. Limestone
    e. Shale

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64. (**2pts**) What chemical composition is typical in making Karst-associated rock have its defined features? Circle all common choices.

a. Magnesium Carbonate
b. Calcium Carbonate
c. Calcium Sulfate
d. Silicon Dioxide
e. Lithium Bicarbonate
f. Ferric Oxide

65. (**1pt**) In areas of Karst, what does a disappearing stream indicate?

a. The stream is evaporating faster than it is supplied
b. The stream is getting incorporated into the grounder faster than it is supplied
c. The stream becomes a subterranean stream with an associated cavern
d. None of the above

66. (**1pt**) Some wetlands can be either primarily rain-fed (ombrotrophic) or groundwater-fed (minerotrophic). One type of these wetlands is more acidic than the other. Which one, and why?

a. Ombrotrophic, because groundwater would have dissolved minerals in it to make it less acidic than rainwater, thereby leaving rainwater as more acidic
b. Minerotrophic, because groundwater could have picked up acidic contaminants to make the water more acidic
c. Ombrotrophic, because the rain input process has stagnant water, which promotes acidity, while the groundwater input process would have moving water, which decreases acidity
d. Minerotrophic, because rainwater ends up being basic due to absorbing carbon dioxide in the air to become carbonate, thus groundwater ends up being more acidic
e. Neither, this is a trick question, and both sources of water end up equally acidic.

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67. (**1pt**) What is the term used to describe the movement of water underground from the vadose zone down to the phreatic zone of an aquifer system?

a. Infiltration
b. Sedimentation
c. Percolation
d. Seepage

68. (**1pt**) As water moves down in the vadose zone toward the phreatic zone, sometimes, there is a slight decrease in the hydraulic conductivity of the sediment. What does this do?

a. The decrease in permeability hinders movement of water downward and creates a new phreatic zone above the original one.
b. The increase in permeability allows water to move downward faster, and creates a dryer layer in the vadose zone above the phreatic zone.
c. The decrease in permeability hinders movement of water downward, and some of it directs laterally, enabling interflow in the vadose zone.
d. The increase in permeability allows water to move downward faster, creating a relatively devoid space of water, sucking in more water in the vadose zone through capillary action, resulting in lateral interflow.

69. (**2pts**) Bert gets lost in a canyon during a hike on vacation. As he was running out of water trying to find a way out, he miraculously stumbles upon water leaking out of one face of the canyon along a series of fissures extending horizontally several tens of meters. What did he find? Select all answers that describe what Bert encountered.

a. A spring
b. A (non-artesian) well
c. An artesian well
d. A confined aquifer
e. An unconfined aquifer
f. A perched aquifer

70. (**1.5pts**) An excavation team digs a hole and finds that water suddenly starts gushing upward out of the hole and above the surface of the ground. *Exactly* what feature did the excavation team dig?

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Figure 1 (Image is flipped sideways)



Remember, do not write on this test document!

Figure 2

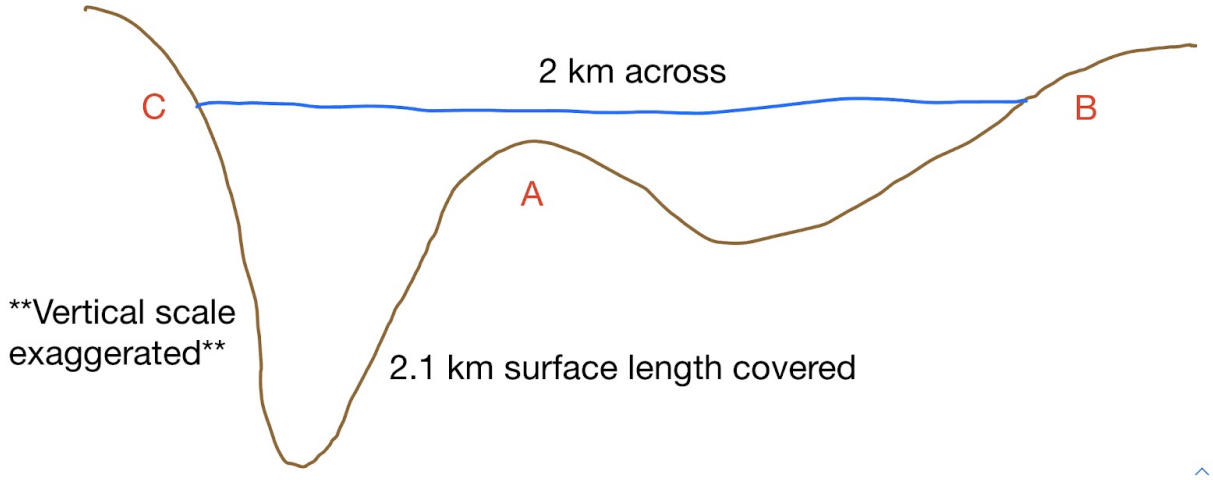


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Figure 3



Figure 4



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Figure 5



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Figure 6

